

Exercise 8.6: The Method of Halves

From a half-angle identity to Viète's infinite product

A step-by-step tutorial for a high-school reader

What you will learn

By the end of this tutorial, you will be able to:

- understand what an infinite product means;
- repeatedly apply the identity $\sin x = 2 \sin(x/2) \cos(x/2)$;
- prove

$$\frac{\sin x}{x} = \prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right);$$

- use $x = \pi/2$ to obtain the famous nested-radical formula for $2/\pi$.

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1 The exercise in our own words

The problem

Starting from the half-angle identity

$$\sin x = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right),$$

prove the infinite-product identity

$$\frac{\sin x}{x} = \prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right).$$

Then put $x = \pi/2$ and show that

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \cdots$$

The phrase *method of halves* means that we keep replacing an angle by half of itself:

$$x, \quad \frac{x}{2}, \quad \frac{x}{4}, \quad \frac{x}{8}, \quad \dots$$

Each halving creates one new cosine factor. The whole proof is built from this simple repeating pattern.

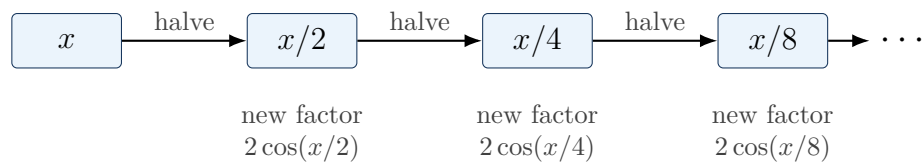


Figure 1. Halve the angle; each new angle contributes one new cosine factor.

2 Background ideas

2.1 Angles must be measured in radians

The proof uses the fundamental limit

$$\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1.$$

This statement is true in this form only when u is measured in *radians*. On a unit circle, an angle of u radians cuts off an arc of length u . Thus radians connect angles directly to lengths and areas, which is why they naturally appear in limits.

2.2 What does the product symbol mean?

The capital Greek letter $\prod$ means “multiply a list of terms.” For example,

$$\prod_{k=1}^4 a_k = a_1 a_2 a_3 a_4.$$

Therefore,

$$\prod_{k=1}^4 \cos\left(\frac{x}{2^k}\right) = \cos\left(\frac{x}{2}\right) \cos\left(\frac{x}{4}\right) \cos\left(\frac{x}{8}\right) \cos\left(\frac{x}{16}\right).$$

Infinite product

An infinite product is defined through its *partial products*. If

$$P_n = \prod_{k=1}^n a_k,$$

and the numbers P_n approach a limit L , then we write

$$\prod_{k=1}^{\infty} a_k = L.$$

Infinity is not a final index that we reach. It means that we study what happens as the number of factors n becomes larger and larger.

2.3 The two half-angle identities we need

For sine, we are given

$$\sin x = 2 \sin(x/2) \cos(x/2).$$

Later we will also use the cosine half-angle identity

$$\cos(\theta/2) = \sqrt{\frac{1 + \cos \theta}{2}}$$

for the small positive angles in this exercise. The positive square root is chosen because all those cosine values are positive.

3 Repeat the sine half-angle formula

We first work with only a *finite* number of halvings. This avoids any mystery about infinity.

3.1 One halving

The given identity says

$$\sin x = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right).$$

3.2 Two halvings

Apply the same identity to $\sin(x/2)$:

$$\sin\left(\frac{x}{2}\right) = 2 \sin\left(\frac{x}{4}\right) \cos\left(\frac{x}{4}\right).$$

Substitute this into the first equation:

$$\begin{aligned} \sin x &= 2 \left[2 \sin\left(\frac{x}{4}\right) \cos\left(\frac{x}{4}\right) \right] \cos\left(\frac{x}{2}\right) \\ &= 2^2 \sin\left(\frac{x}{4}\right) \cos\left(\frac{x}{2}\right) \cos\left(\frac{x}{4}\right). \end{aligned}$$

3.3 Three halvings

Now apply the formula to $\sin(x/4)$:

$$\sin\left(\frac{x}{4}\right) = 2 \sin\left(\frac{x}{8}\right) \cos\left(\frac{x}{8}\right).$$

This gives

$$\sin x = 2^3 \sin\left(\frac{x}{8}\right) \cos\left(\frac{x}{2}\right) \cos\left(\frac{x}{4}\right) \cos\left(\frac{x}{8}\right).$$

The pattern is now visible: after n halvings, there is one small sine term, a factor 2^n , and n cosine factors.

Finite repeated-halving identity

For every positive integer n ,

$$\sin x = 2^n \sin\left(\frac{x}{2^n}\right) \prod_{k=1}^n \cos\left(\frac{x}{2^k}\right). \quad (1)$$

3.4 Why formula (1) is always true: a short induction

A proof by induction is a formal way to confirm that the pattern continues forever.

Base case. For $n = 1$, formula (1) is exactly the given half-angle identity.

Induction step. Suppose formula (1) is true for some n . Then

$$\sin\left(\frac{x}{2^n}\right) = 2 \sin\left(\frac{x}{2^{n+1}}\right) \cos\left(\frac{x}{2^{n+1}}\right).$$

Substituting this into (1) adds one factor of 2 and one new cosine factor:

$$\begin{aligned} \sin x &= 2^{n+1} \sin\left(\frac{x}{2^{n+1}}\right) \left(\prod_{k=1}^n \cos\left(\frac{x}{2^k}\right) \right) \cos\left(\frac{x}{2^{n+1}}\right) \\ &= 2^{n+1} \sin\left(\frac{x}{2^{n+1}}\right) \prod_{k=1}^{n+1} \cos\left(\frac{x}{2^k}\right). \end{aligned}$$

So the formula is true for $n + 1$, completing the induction.

4 Divide by x and reveal the small-angle ratio

Assume first that $x \neq 0$. Divide equation (1) by x :

$$\frac{\sin x}{x} = \frac{2^n \sin(x/2^n)}{x} \prod_{k=1}^n \cos\left(\frac{x}{2^k}\right).$$

The first fraction can be rewritten as

$$\frac{2^n \sin(x/2^n)}{x} = \frac{\sin(x/2^n)}{x/2^n}.$$

Therefore,

The key finite formula

$$\frac{\sin x}{x} = \frac{\sin(x/2^n)}{x/2^n} \prod_{k=1}^n \cos\left(\frac{x}{2^k}\right). \quad (2)$$

This equation separates the problem into two pieces:

1. the small-angle factor

$$\frac{\sin(x/2^n)}{x/2^n};$$

2. the partial product

$$P_n = \prod_{k=1}^n \cos\left(\frac{x}{2^k}\right).$$

As n grows, the angle $x/2^n$ approaches 0. We therefore need to understand $\sin u/u$ for very small u .

5 Why $\sin u/u$ approaches 1

This important limit can be understood geometrically. Take a unit circle and an angle u with $0 < u < \pi/2$.

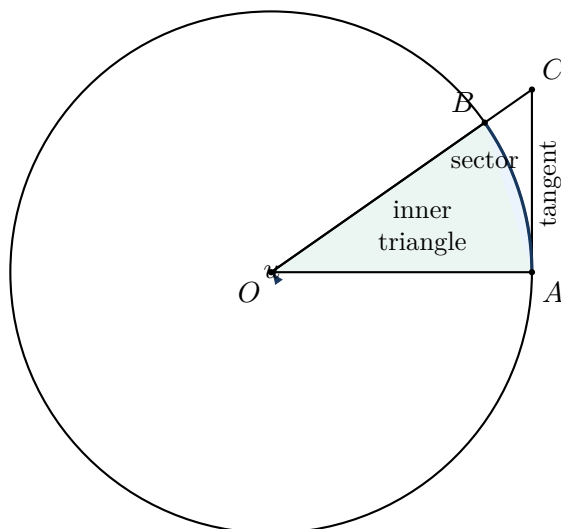


Figure 2. The inner triangle lies inside the sector, which lies inside the outer triangle.

For a unit circle:

- the inner triangle OAB has area $\frac{1}{2} \sin u$;
- the circular sector has area $\frac{1}{2}u$ because u is in radians;
- the outer triangle OAC has area $\frac{1}{2} \tan u$.

Since the three regions are nested,

$$\frac{1}{2} \sin u < \frac{1}{2}u < \frac{1}{2} \tan u.$$

Multiply by 2:

$$\sin u < u < \tan u.$$

The left inequality gives

$$\frac{\sin u}{u} < 1.$$

Using $\tan u = \sin u / \cos u$, the right inequality gives

$$u < \frac{\sin u}{\cos u} \implies \cos u < \frac{\sin u}{u}.$$

Thus

$$\cos u < \frac{\sin u}{u} < 1.$$

When $u \rightarrow 0$, we have $\cos u \rightarrow 1$. The ratio is squeezed between two quantities that both approach 1, so

Small-angle limit

$$\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1. \quad (3)$$

For negative u , the same conclusion follows because

$$\frac{\sin(-u)}{-u} = \frac{\sin u}{u}.$$

6 Pass from the finite product to the infinite product

Let

$$u_n = \frac{x}{2^n}.$$

Because the denominator 2^n grows without bound,

$$u_n \longrightarrow 0.$$

By the small-angle limit (3),

$$\frac{\sin(x/2^n)}{x/2^n} \longrightarrow 1.$$

Now solve equation (2) for the partial product:

$$P_n = \prod_{k=1}^n \cos\left(\frac{x}{2^k}\right) = \frac{\sin x/x}{\sin(x/2^n)/(x/2^n)}.$$

As $n \rightarrow \infty$, the denominator on the right approaches 1. Hence

$$P_n \rightarrow \frac{\sin x}{x}.$$

By the definition of an infinite product,

First requested identity

For $x \neq 0$,

$$\frac{\sin x}{x} = \prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right). \quad (4)$$

At $x = 0$, the left side is understood by its continuous extension as 1, and every factor on the right is $\cos 0 = 1$, so the identity remains valid in the extended sense.

What really happened?

The repeated-halving formula left behind a small sine:

$$\sin(x/2^n).$$

After division by x , this became the ratio

$$\frac{\sin(x/2^n)}{x/2^n},$$

which becomes almost exactly 1 when n is large. The remaining cosine factors therefore carry almost all of the value of $\sin x/x$.

7 Choose $x = \pi/2$

Now substitute $x = \pi/2$ into (4). The left side is

$$\frac{\sin(\pi/2)}{\pi/2} = \frac{1}{\pi/2} = \frac{2}{\pi}.$$

The k -th cosine factor becomes

$$\cos\left(\frac{\pi/2}{2^k}\right) = \cos\left(\frac{\pi}{2^{k+1}}\right).$$

Therefore,

Product for $2/\pi$

$$\frac{2}{\pi} = \prod_{k=1}^{\infty} \cos\left(\frac{\pi}{2^{k+1}}\right) = \cos \frac{\pi}{4} \cos \frac{\pi}{8} \cos \frac{\pi}{16} \cdots. \quad (5)$$

The rest of the exercise is to rewrite these cosine values as nested square roots.

8 Turn the cosine factors into nested radicals

8.1 Derive the cosine half-angle formula

The double-angle identity

$$\cos(2t) = 2 \cos^2 t - 1$$

can be rearranged:

$$2 \cos^2 t = 1 + \cos(2t),$$

so

$$\cos^2 t = \frac{1 + \cos(2t)}{2}.$$

Taking the positive square root for the angles used here gives

$$\cos t = \sqrt{\frac{1 + \cos(2t)}{2}}.$$

Writing $t = \theta/2$, we get

$$\boxed{\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1 + \cos \theta}{2}}.} \quad (6)$$

8.2 Compute the first factor

Use $\theta = \pi/2$:

$$\begin{aligned} \cos \frac{\pi}{4} &= \sqrt{\frac{1 + \cos(\pi/2)}{2}} \\ &= \sqrt{\frac{1 + 0}{2}} \\ &= \frac{\sqrt{2}}{2}. \end{aligned}$$

8.3 Compute the second factor

Use $\theta = \pi/4$:

$$\begin{aligned} \cos \frac{\pi}{8} &= \sqrt{\frac{1 + \cos(\pi/4)}{2}} \\ &= \sqrt{\frac{1 + \sqrt{2}/2}{2}} \\ &= \sqrt{\frac{2 + \sqrt{2}}{4}} \\ &= \frac{\sqrt{2 + \sqrt{2}}}{2}. \end{aligned}$$

8.4 Compute the third factor

Use $\theta = \pi/8$:

$$\begin{aligned} \cos \frac{\pi}{16} &= \sqrt{\frac{1 + \cos(\pi/8)}{2}} \\ &= \sqrt{\frac{1 + \frac{1}{2}\sqrt{2 + \sqrt{2}}}{2}} \\ &= \frac{\sqrt{2 + \sqrt{2 + \sqrt{2}}}}{2}. \end{aligned}$$

Each new radical is obtained by placing the previous radical after another 2+ and then taking a square root.

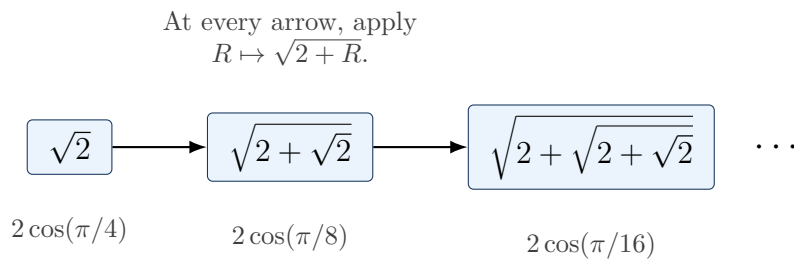


Figure 3. The nested radicals follow the same repeated-halving pattern as the angles.

8.5 A clean recursive description

Define

$$R_1 = \sqrt{2}, \quad R_{n+1} = \sqrt{2 + R_n}.$$

Then

$$R_n = 2 \cos\left(\frac{\pi}{2^{n+1}}\right).$$

To see why, check the first value:

$$R_1 = \sqrt{2} = 2 \cos \frac{\pi}{4}.$$

If $R_n = 2 \cos \theta$, then

$$\begin{aligned} R_{n+1} &= \sqrt{2 + 2 \cos \theta} \\ &= \sqrt{4 \cos^2(\theta/2)} \\ &= 2 \cos \frac{\theta}{2}, \end{aligned}$$

where the last cosine is positive. Thus one radical step corresponds exactly to halving the angle.

Substituting the radical forms into equation (5) gives the requested result.

Second requested identity: Viète's formula

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \cdots \quad (7)$$

This is commonly called Viète's product for $2/\pi$.

9 A numerical check

An infinite identity becomes easier to trust when its first few partial products move toward the claimed value. Let

$$V_n = \prod_{k=1}^n \cos\left(\frac{\pi}{2^{k+1}}\right).$$

The target value is

$$\frac{2}{\pi} \approx 0.6366197724.$$

n	New factor	Partial product V_n
1	$\cos(\pi/4)$	0.7071067812
2	$\cos(\pi/8)$	0.6532814824
3	$\cos(\pi/16)$	0.6407288619
4	$\cos(\pi/32)$	0.6376435773
5	$\cos(\pi/64)$	0.6368755077
6	$\cos(\pi/128)$	0.6366836927

The values decrease toward 0.6366197724, exactly as the formula predicts. Later factors are very close to 1, but multiplying many of them still makes a small correction.

10 A concise proof suitable for a homework submission

The following is a compact version of the argument after you understand all the steps.

Complete proof

Repeated use of

$$\sin t = 2 \sin(t/2) \cos(t/2)$$

gives, for every $n \geq 1$,

$$\sin x = 2^n \sin\left(\frac{x}{2^n}\right) \prod_{k=1}^n \cos\left(\frac{x}{2^k}\right).$$

For $x \neq 0$, division by x yields

$$\frac{\sin x}{x} = \frac{\sin(x/2^n)}{x/2^n} \prod_{k=1}^n \cos\left(\frac{x}{2^k}\right).$$

Since $x/2^n \rightarrow 0$ and $\lim_{u \rightarrow 0} \sin u/u = 1$, taking $n \rightarrow \infty$ gives

$$\frac{\sin x}{x} = \prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right).$$

Now set $x = \pi/2$. Since $\sin(\pi/2) = 1$,

$$\frac{2}{\pi} = \cos \frac{\pi}{4} \cos \frac{\pi}{8} \cos \frac{\pi}{16} \cdots.$$

Using

$$\cos(\theta/2) = \sqrt{\frac{1 + \cos \theta}{2}},$$

we obtain

$$\cos \frac{\pi}{4} = \frac{\sqrt{2}}{2},$$

$$\cos \frac{\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2},$$

and

$$\cos \frac{\pi}{16} = \frac{\sqrt{2 + \sqrt{2 + \sqrt{2}}}}{2},$$

with the same pattern continuing. Therefore,

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2 + \sqrt{2}}}{2} \cdot \frac{\sqrt{2 + \sqrt{2 + \sqrt{2}}}}{2} \cdots.$$

11 Common mistakes and how to avoid them

Mistake 1: losing powers of 2

Every halving introduces a factor of 2. After n halvings, the coefficient is 2^n , not merely $2n$ or 2.

Mistake 2: an index shift

After setting $x = \pi/2$,

$$\frac{x}{2^k} = \frac{\pi}{2^{k+1}}.$$

Thus the first factor is $\cos(\pi/4)$, not $\cos(\pi/2)$.

Mistake 3: treating infinity as a final step

We never “plug in $n = \infty$.” We form the finite partial products P_n and then take the limit as $n \rightarrow \infty$.

Mistake 4: choosing the wrong square root

From $z^2 = A$, one normally has $z = \pm\sqrt{A}$. Here all angles $\pi/4, \pi/8, \dots$ lie between 0 and $\pi/2$, where cosine is positive. Therefore we use the positive root.

Mistake 5: using degrees in the small-angle limit

The limit $\sin u/u \rightarrow 1$ requires radian measure. In degrees, a conversion factor would appear.

12 Check your understanding

Try these before reading the answers.

1. Write the repeated-halving identity after exactly four halvings.
2. Explain in one sentence why $x/2^n \rightarrow 0$.
3. What is the fourth cosine factor after setting $x = \pi/2$?
4. Let $R_1 = \sqrt{2}$ and $R_{n+1} = \sqrt{2 + R_n}$. Write R_4 .
5. Why is the positive square root used in the cosine half-angle formula here?

Answers.

1.

$$\sin x = 2^4 \sin \frac{x}{16} \cos \frac{x}{2} \cos \frac{x}{4} \cos \frac{x}{8} \cos \frac{x}{16}.$$

2. The numerator x stays fixed while the positive denominator 2^n grows without bound.3. $\cos(\pi/32)$, because the k -th factor is $\cos(\pi/2^{k+1})$.

4.

$$R_4 = \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2}}}}.$$

5. All relevant angles lie in the first quadrant, where cosine is positive.

13 Final summary

The proof has one repeating mechanism and one limiting step:

halve the angle repeatedly $\Downarrow$ $\sin x = 2^n \sin(x/2^n) \prod_{k=1}^n \cos(x/2^k)$ $\Downarrow$ divide by x $\frac{\sin x}{x} = \frac{\sin(x/2^n)}{x/2^n} \prod_{k=1}^n \cos(x/2^k)$ $\Downarrow n \rightarrow \infty, \frac{\sin u}{u} \rightarrow 1$ $\frac{\sin x}{x} = \prod_{k=1}^{\infty} \cos(x/2^k).$

Choosing $x = \pi/2$ changes the left side into $2/\pi$. Repeated use of the cosine half-angle identity changes the factors on the right into nested radicals. That is Viète's elegant product.